

Roberto (André) Mossi Milla

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SUMMARY

Software Engineer and published AI researcher experienced in artificial intelligence, computer vision, evolutionary computation, full-stack systems, and real-time simulation. Background spans academic research, industrial computer vision, multimodal AI, AI-assisted healthcare systems, backend architecture, independent software development, and end-to-end engineering. Interested in developing intelligent systems that combine learning, perception, optimization, and scalable software infrastructure.

EDUCATION

Gannon University, Pennsylvania, United States

Bachelor of Science in Computer Systems Engineering, May 2025[11]

ISI Japanese Language School, Tokyo, Japan

Intensive Japanese Language Program, October 2025 - March 2026[9]

Harvard University, Online

CS50's Introduction to Computer Science, December 2022[12]

Extreme Networks, Online

Introduction to Future Networks, May 2022[1]

RESEARCH EXPERIENCE

Gannon University, Pennsylvania, United States

Research Assistant, August 2024 - May 2025

- Conducted faculty-supervised research on multi-agent artificial intelligence, investigating natural selection, survival, adaptation, and emergent behavior in dynamic three-dimensional environments.[10]
- Independently implemented NeuroEvolution of Augmenting Topologies (NEAT) in C++, including the genome representation used to encode neural-network structures and behavioral strategies across generations.
- Developed a 3D multi-agent simulation environment in Unreal Engine featuring procedural world generation, physics-based interactions, autonomous agents, and evolutionary experimentation using C++, and CUDA
- Designed procedurally generated environments containing halls, floors, walls, and props, allowing experiments across randomized scenarios without manually constructing each environment.
- Used CUDA to support GPU-accelerated simulation and computational workloads, separating processing from rendering to improve the scalability of experiments.
- Research resulted in a peer-reviewed publication through the American Society for Engineering Education (ASEE) and a research presentation at a Sigma Xi conference at Penn State University.[6] [3]

INDUSTRY EXPERIENCE

Ingeniería Digital, Tegucigalpa, Honduras

Full-Stack Developer / AI Systems Engineer, May 2026 - Present

AI-Assisted Medical Document Generation Platform

- Built an AI-assisted medical document generation platform from the ground up for post-surgery reports (IDR), reducing report preparation from approximately one week to 1–2 days and reducing final document generation to under one minute.
- Combined OCR extraction, LLM-assisted grammar correction, and automated professional document structuring using Next.js, React, TypeScript, PostgreSQL, AWS S3, Amazon Textract, and OpenAI APIs.
- Architected authenticated document-management workflows using Auth0, enabling users to upload, generate, analyze, edit, version, review, and manage AI-assisted reports.
- Designed file-processing pipelines integrating OCR extraction, cloud storage, PostgreSQL persistence, document retrieval, and structured processing workflows.
- Implemented internationalization workflows and automated translation-asset generation using CMake to support multilingual deployments across US and Honduran clients.

AI Vehicle Damage Assessment Platform

- Developed an AI-assisted vehicle damage assessment platform that reduced assessment workflows from more than one week to approximately a minute while automatically generating insurance estimates.
- Integrated AWS Rekognition to localize vehicles within uploaded images and Gemini Flash to detect and analyze vehicle damage within the identified vehicle boundaries.
- Integrated multimodal AI, computer vision, VIN-history retrieval, and insurance-estimation workflows to support vehicle valuation and claims analysis.
- Developed data-acquisition services using Playwright to collect, normalize, and process vehicle information and historical data from external sources.
- Implemented interactive damage annotation and review tools using Konva, allowing users and insurance personnel to identify, move, visualize, and validate detected damage regions.
- Developed an experimental AI evaluation workflow using persisted visual annotations, enabling evaluation of different models, processing stages, and combinations of approaches.

Gannon University, Pennsylvania, United States

Computer Vision Engineer (Project-Based) Industry Partner (NDA), August 2024 - May 2025

- Architected and deployed components of a real-time industrial traceability system using LiDAR-triggered image capture and Ultralytics YOLOv8 to detect and track packages and pallets, automatically generating structured events with images and detection metadata.[4]
- Collaborated across hardware, backend, and software components while taking primary ownership of the monitoring dashboard and visualization interface.
- Developed a full-stack web dashboard integrated with backend services and TimescaleDB to visualize detection events, images, and operational information in near real time.
- Implemented administrative tools to curate datasets, move and correct bounding boxes, and improve iterative object-detection model retraining workflows.
- Developed image-quality and preprocessing workflows using contrast, color, filtering, spatial image partitioning, and reconstruction techniques to improve image usability.
- Engineered an event-driven pipeline synchronizing frame capture with detection metadata, enabling downstream anomaly and damage-analysis workflows.
- Containerized distributed services using Docker and integrated multi-language components (C++, Python, Rust) to coordinate inference, data processing, and system communication.

Dinant, Tegucigalpa, Honduras

Full-Stack Developer, June - July 2023

- Led the end-to-end design and development of two production web platforms, conducting requirements gathering, architecture design, client presentations, and deployment.[2]
- Architected a contract lifecycle management platform supporting device sales, rentals, repairs, and multi-country operations using relational SQL data models.
- Developed a multi-role web application using Oracle APEX, JavaScript, HTML, and CSS supporting digital signatures, PDF generation, and administrative reporting.
- Built a factory maintenance management platform enabling machine monitoring, maintenance workflows, repair tracking, inventory reporting, and procurement insights.

PROJECTS & ENTREPRENEURSHIP

IRIS, Founder, Product Designer & Software Engineer — 2026 - Present

- Founded and lead development of IRIS, an AI-powered computer vision platform designed to provide accessible vision intelligence through a scalable API and user-facing software platform.
- Architected cloud-based computer-vision infrastructure supporting authenticated API access, image-processing workflows, model inference, and scalable interaction between client applications and AI services.
- Designed multimodal AI workflows combining computer-vision models and large language models to transform visual inputs into structured, actionable information.
- Developed secure API authentication, user-specific access tokens, asynchronous processing, and scalable backend services for integrating computer-vision capabilities into external applications.
- Designed the platform as both a developer-facing API and consumer-facing application, allowing external software to integrate computer-vision capabilities.

NOIR, Founder, Product Designer & Software Engineer — 2025 - Present

- Founded and lead development of NOIR, a cross-platform movie discovery and continuity platform spanning streaming services, theaters, and devices.
- Architected a distributed backend using Rust, SeaORM, Axum, PostgreSQL, Redis, Supabase, Docker, and Railway for authentication, metadata aggregation, availability services, synchronization, and persistent user state.
- Developed native SwiftUI applications targeting macOS, iOS, iPadOS, and watchOS, with cross-device synchronization and AI-assisted natural-language movie discovery.
- Designed an AI intelligence layer translating natural-language viewing intent into structured movie discovery and availability queries.

Copilot-Lualine, Open Source Project

- Created and maintain Copilot-Lualine, an open-source Neovim plugin integrating GitHub Copilot status and controls into the Lualine statusline ecosystem; the project has received 117+ GitHub stars.[5]
- Implemented Lua-based editor tooling focused on workflow visibility, developer productivity, and Neovim customization.

LEADERSHIP & ORGANIZATIONS

- Founder, Product Designer & Software Engineer, NOIR (2025 – Present).
- Founder, Product Designer & Software Engineer, IRIS (2026 – Present).
- Member, Mathematics Club, Gannon University (2023 - 2025).
- Member, Computer Club, Gannon University (2022 - 2025).
- President, Computer Club, Dowal School (High School).
- Member, Leo Club, Dowal School (High School).

PUBLICATIONS

- Mossi, R. A. — “Tag AI-Sandbox,” Proceedings of the ASEE Annual Conference, 2025 (peer-reviewed).[6]

TECHNICAL SKILLS

- Languages: C++, Rust, Swift, TypeScript, JavaScript, Python, Java, SQL, Lua, Bash, PowerShell.
- Frontend: SwiftUI, React, Next.js, Vite, Tailwind CSS, Shadcn/UI, Konva.
- Backend & Cloud: Axum, SeaORM, Tokio, PostgreSQL, Redis, Supabase, AWS S3, Amazon Textract, Docker, Auth0, Railway, REST APIs.
- AI & Computer Vision: OpenAI APIs, Gemini, Claude APIs, YOLOv8, CUDA, PyTorch, NEAT, evolutionary computation, OCR pipelines, multimodal AI workflows.
- Developer Tooling: GitHub Actions, Husky, ESLint, Prettier, Commitizen, CMake, SwiftLint, Clippy, Git.
- Operating Systems: Linux, macOS, Windows.
- Languages: Spanish (Native), English (Fluent), Japanese (Intermediate).

AWARDS AND HONORS

- “School Development Award” Award obtained for outstanding development of a videogame using Blueprints in Unreal Engine 4, 2020[8]
- “School Development Award” Award obtained for outstanding development of a videogame using C++ and Unreal Engine 4, 2016[7]

REFERENCES

- [1] Extreme Academy. *Extreme Networks Certification*. Dec. 1, 2022. URL: <https://andremossi.vercel.app/experiences/extreme> (visited on 02/15/2026).
- [2] Rigoberto Funes. *Dinant Recommendation Letter*. Feb. 11, 2026. URL: <https://andremossi.vercel.app/PDF/Recommendations/dinant.pdf> (visited on 02/15/2026).
- [3] lah585. *Tag AI-Sandbox Presentation*. Sigma Xi. Mar. 11, 2025. URL: <https://sites.psu.edu/behrendsigmaxi2025/2025/03/11/tag-ai-sandbox/> (visited on 02/15/2026).
- [4] Andre Mossi. *Computer Vision Project*. Feb. 15, 2026. URL: <https://andremossi.vercel.app/experiences/traceability> (visited on 02/15/2026).
- [5] Andre Mossi. *Copilot Lualine*. Nov. 28, 2023. URL: <https://github.com/AndreM222/copilot-lualine> (visited on 08/26/2026).
- [6] Roberto Andre Mossi and Ramakrishnan Sundaram. "Tag AI-Sandbox Paper". In: *2025 ASEE North Central Section (NCS) Annual Conference*. 10.18260/1-2-54692. Marshall University, Huntington, West Virginia: ASEE Conferences, Mar. 2025. URL: <https://peer.asee.org/54692>.
- [7] Dowal School. *Software Engineer 2016 Award*. May 12, 2016. URL: <https://andremossi.vercel.app/experiences/dowal2016> (visited on 02/15/2026).
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- [9] ISI Japanese Language School. *ISI Japanese School*. Mar. 10, 2026. URL: <https://andremossi.vercel.app/experiences/isi> (visited on 06/02/2026).
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- [12] Harvard University. *Harvard CS50 Certification*. Dec. 20, 2022. URL: <https://andremossi.vercel.app/experiences/harvard> (visited on 02/15/2026).